

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
11	22	33	44	55	66	77	88	99

Left
Mid
Right

1) Key = 33.

2) Find Left : 0

Right : 8

Mid = 4

3) Key == arr[mid]

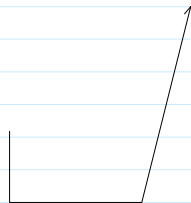
33 == 55 ? No

4) Check if the key is smaller to the mid element or greater.

As the key is smaller , we will continue the search in the Left Sub array

LSA : Left to Mid-1

0 to 3



LSA RSA

[0]	[1]	[2]	[3]
11	22	33	44

Left Mid Right

LSA → left
to
m-1

RSA → m + 1
to
Right

Continue with steps 2, 3 and 4

2) Left : 0

Right : Mid - 1 : 3

Mid : (0+3) / 2 = 1

3) Compare the key with the mid element.

33 == arr[mid]

33 == 22 ? NO

4) Check if 33 is smaller or greater.

As 33 is greater continue the search in RSA.

RSA is from Mid + 1 to Right.

2 to 3

[2]	[3]
33	44

Left Right
Mid

Continue steps 2 and 3

2) Left = mid+1 = 2

Right = 3

Mid = (2+3) / 2 = 2

3) Compare the key with the mid element.

33 == arr[mid]

33 == 33 ? YES

Key found at index 2